# Multiview Multitask Gaze Estimation With Deep Convolutional Neural Networks

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***Abstract*— Gaze estimation, which aims to predict gaze points with given eye images, is an important task in computer vision because of its applications in human visual attention understanding. Many existing methods are based on a single camera, and most of them only focus on either the gaze point estimation or gaze direction estimation. In this paper, we propose a novel multitask method for the gaze point estimation using multiview cameras. Specifically, we analyze the close relationship between the gaze point estimation and gaze direction estimation, and we use a partially shared convolutional neural networks architecture to simultaneously estimate the gaze direction and gaze point. Furthermore, we also introduce a new multiview gaze tracking data set that consists of multiview eye images of different subjects. As far as we know, it is the largest multiview gaze tracking data set. Comprehensive experiments on our multiview gaze tracking data set and existing data sets demonstrate that our multiview multitask gaze point estimation solution consistently outperforms existing methods.**

***Index Terms*— Convolutional neural networks (CNNs), gaze tracking, multitask learning (MTL), multiview learning.**

## I. Introduction

GAZE tracking or gaze estimation is an important topic for understanding human visual attention. Such a technology has been widely deployed in various fields, such as human–computer interaction [1]–[3], visual behavior analysis [3], and psychological studies [4]. Based on its huge potential value, many eye tracking devices (e.g., Tobii X3-120, Tobii EyeX, and Eye Tribe [5]) have come into being. However, most of these devices are very expensive to purchase, making them hampered in wide adoption. Usually, appearance-based gaze estimation employs a top-down strategy [6], which predicts the gaze direction or gaze point through eye images directly. Such an approach is well established as another alternative for eye tracking since only achieving eye images is much cheaper.

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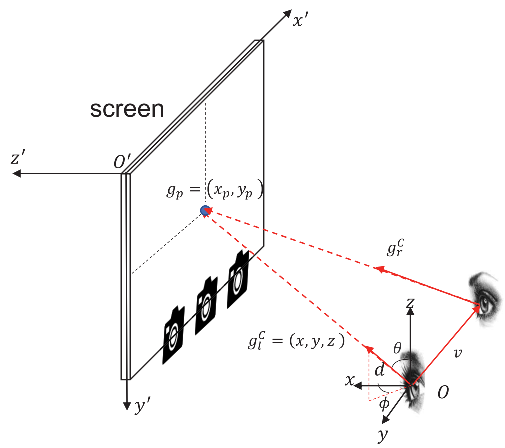



Fig. 1. Gaze point position depends on the gaze direction and eye position. $O$ is the eye position in the 3-D coordinate system, $g_l^C$ and $g_r^C$ are the gaze directions of the left eye and right eye, respectively, $g_p$ is the 2-D gaze point, and $v$ is the vector from left eye to right eye.

For appearance-based gaze tracking, many previous works [5], [7] attempt to collect data using a single-view camera. Most of them either fix the head poses or ignore depth information between the eyes of different subjects and the screen target. As shown in Fig. 1, both eye position $O$ in a 3-D coordinate system and gaze direction $g_l^C$ determine the gaze point position $g_p$ in the 2-D plane. Given the same gaze direction, even a slight depth change of eyes may easily introduce a large estimation bias. Whereas, a single-view camera cannot estimate the depth of the eye well. Consequently, subjects are constrained to adjust their head poses in the most single-view gaze tracking systems to make sure that their eyes can be captured by the camera and fix the depth. However, such a constraint will harm the user experience. Fortunately, the depth between the eyes and the screen can be inferred from multiview images. Thus, we propose to leverage images captured by multiview cameras for better gaze tracking, and such a multiview camera system introduces depth information for gaze tracking indirectly. Previous works [8], [9] have shown that the multiview learning would leverage data from different views to learn a more robust representation with a better generalization. In the gaze tracking application, compared with single-view-based gaze tracking, multiview cameras relax the constraint on head poses of participants. Even if the captured eye images in one view contain some self-occlusion cases caused by the head pose, it is possible that the eye images are captured properly by some other views. In addition, it can reduce the effect of specularity and color distortion caused by eyeglasses, which makes it more robust and easier to be deployed in a real environment. At present, more and more mobile phones are deployed with multiple camera sensors (for example, iPhone 7 Plus and Huawei P9), which also provides the basis and the application scenarios for the research on multiview gaze tracking. However, there is only one multiview gaze tracking data set in this community (the UT Multiview data set [10]), which is captured with only 50 subjects and the head pose is fixed. Such a small data set restricts the exploration of data-driven-based methods for this task. This motivates us to build a new large multiview gaze tracking data set with the flexible head pose. Specifically, the number of subjects in our data set is 137, which is much larger than that in UT Multiview [10], and the number of images is almost four times more than that in UT Multiview. Our data set can be downloaded here.[1]

To tackle gaze tracking, previous works [5], [11] estimate either the gaze direction [11] or gaze point alone [5] and do not consider their close relationship. However, as shown in Fig. 1, the gaze direction prediction and gaze estimation prediction are two closely related tasks [12]. Once the gaze direction is known, based on the eye position information, we can infer the coordinates of the gaze point on the screen target by intersecting the estimated 3-D gaze vector with the screen plane. Nevertheless, since the gaze direction is usually not perpendicular to the screen, even the same gaze direction errors still lead to different gaze point prediction errors along different directions [as shown in Fig. 4(a)]. Now that the gaze point prediction depends on the gaze direction prediction, in light of the successes of multitask learning (MTL) in many computer vision tasks [13]–[15], we propose to tackle the gaze point estimation within an MTL framework. MTL can learn better feature representation for different but related tasks by (partially) sharing the network parameters, especially when the training samples are limited [16], [17]. In this paper, we simultaneously estimate the gaze direction and gaze point via a partially shared convolutional neural networks (CNNs), and here, the two tasks are the gaze direction estimation and gaze point estimation, respectively. Furthermore, since the gaze directions of both eyes actually intersect at the same gaze point on the screen, besides making the predicted gaze direction agrees with the ground-truth gaze direction for both eyes, we also impose a coplanar constraint on the estimated gaze direction for both eyes. Since we are tackling the multiview gaze tracking data, and all eyes are roughly aligned already, besides concatenating features from different views, we also introduce an additional cross-view pooling module that conducts elementwise maximization over features at the same location but from different views. Such a cross-view pooling will reduce the effect of information loss caused by self-occlusion and noises in terms of specularity and color distortion in images captured by different views, hence boosts the robustness of gaze tracking.

The contributions of this paper can be summarized as follows: 1) we build a new multiview gaze tracking data set. As far as we know, it is the largest multiview gaze tracking data set in terms of the number of both subjects and images. Such a data set helps to explore the effect of human poses and multiview images on gaze tracking; 2) a multiview MTL framework has been proposed to deal with both the gaze direction estimation and gaze point estimation simultaneously. For the gaze direction prediction, we propose to introduce a coplanar constraint on the gaze directions of left and right eyes. For the gaze point estimation, we propose to introduce a cross-view pooling module. Experimental results validate the effectiveness of our model; and 3) we conduct extensive experiments to analyze the effect of different factors related to gaze tracking, including network architecture, information aggregation strategy for features extracted with different views, number of subjects, and so on. These analyses would facilitate the design of appearance-based gaze tracking system in real applications.

The rest of this paper is organized as follows. Section II reviews the related work. Section III introduces our multiview gaze tracking data set. Section IV presents our multiview multitask gaze tracking method. We empirically evaluate our method in Section V, and conclude our work in Section VI.

## II. Related Work

In this section, we briefly review the existing methods for the gaze direction estimation and gaze point estimation and some existing data sets used for gaze tracking.

### A. Gaze Tracking Tasks

Gaze tracking tasks can be categorized as 2-D gaze point estimation and 3-D gaze direction estimation.

*1) 2-D Gaze Point Estimation:* A 2-D gaze point estimation task is to predict the coordinate of the point on the screen that a person stares at. Early works obtained the eye position with a chin-rest-fixed head pose of a participant [18]–[21]. Such a fixed head pose solution avoids predicting the eye position and tremendously simplifies the gaze point estimation. Then, the 2-D gaze point estimation task was cast as a regression problem from eye images to the gaze point coordinate. Even though this solution eased the complexity of prediction, it restricted the flexibility of gaze tracking in real applications. Recently, Krafka *et al.* [5] and Zhang *et al.* [12] proposed to relax the constraint for the head pose, thereby used a CNN framework and its variants to estimate the gaze point from face images with arbitrary head poses. However, the important depth information between the eyes and the screen target was still not utilized in these works [5], [12], which restricted the accuracy.

*2) 3-D Gaze Direction Estimation:* A 3-D gaze direction estimation task is to predict the direction (usually a unit vector) in which a person is looking [11], [22]–[24]. Unlike the gaze point estimation, it can directly be inferred from eye images without any eye position information. Usually, the eye center position is set as the initial point and the gaze point on the screen target is the endpoint. Then, the ground-truth direction can be denoted as a direction vector from the initial point to the endpoint. Sugano *et al.* [10] proposed a data normalization technique to restrict the appearance variation

[1] https://github.com/dongzelian/multi-view-gaze

into a single, normalized training space, which is demonstrated to be successful for the gaze direction estimation. In [11], a similar method was adopted, and then the normalized eye images and head pose were fed into a multimodal CNN to estimate the gaze direction.

### B. Gaze Estimation Approaches

Usually, gaze estimation can be categorized into model-based approaches and appearance-based approaches [25].

*1) Model-Based Approaches:* Model-based approaches use a geometric eye model and feature for gaze estimation. It can be further divided into corneal-reflection and shape-based methods. Corneal-reflection methods [26]–[30] utilized specialized hardware such as external infrared light sources to extract eye features, which were usually represented as pupil-glint vectors. Then, a gaze mapping function was acquired through the calibration procedure. The gaze point position was computed from the pupil and the Purkinje image features by using the eyeball model in [26]. However, the head position needed to be fixed in this method. In [28], a gaze tracking system with natural head movements was proposed, which analytically obtained a head mapping function to compensate the head movement. The calibration procedure was only employed one time for each individual in this system. In contrast, shape-based methods [20], [31]–[33] directly inferred the gaze direction from observed eye shapes, such as pupil centers or iris edges. A hybrid scheme was proposed to combine the head pose and eye location information to obtain gaze estimation in [20]. However, shape-based approaches might not be robust under variable lighting conditions and hard to process low-resolution eye images. Thus, it could process short distance scenarios well but effectiveness in mid-distance scenes was unclear.

*2) Appearance-Based Approaches:* Appearance-based approaches [18], [19], [22], [34], [35] usually extracted a high-dimensional vector from a single-eye image as the feature and learned a mapping from the feature to the gaze point or gaze direction. Compared to model-based methods, appearance-based methods more potentially realized a nonintrusive, calibration-free gaze estimation task and worked with low-resolution eye images.

In appearance-based gaze estimation methods, the mapping function from the feature to gaze point (or gaze direction) could be learned in multiple ways. Lu *et al.* [36] proposed a method to adaptively find the subset of training samples to linearly represent the test images. Later, they proposed to add the blink detection and alignment of the eye images to increase the accuracy [22]. Sugano *et al.* [6] treated the saliency maps as the probability distribution maps to predict the gaze points by using the Gaussian process regression. Different from those works, Baluja and Pomerleau [18] first developed an artificial neural network-based gaze tracking system without the need of wearing any special equipment. It mimicked and learned the way how humans looked at points. Besides, Sewell and Komogortsev [34] realized a real-time gaze tracking system through a standard webcam.

Recently, deep learning methods have been applied to a variety of tasks in computer vision and achieve great successes, such as image classification [37], [38], semantic segmentation [39], [40], and video analysis [41], [42]. Compared to some conventional feature representation methods [43], [44], deep learning method extracts more robust features. Similarly, some works used the CNN to estimate the gaze point (2-D) or gaze direction (3-D) directly [5], [11]. Zhang *et al.* [11] utilized the head pose as extra information and fed eye images and head pose into the multimodal CNN for the gaze direction estimation. Krafka *et al.* [5] indicated that a multiregion CNN architecture that took both eyes and face images as input and the gaze point as output could benefit the gaze estimation performance. In [12], the spatial weights CNN was learned for the gaze point estimation. Furthermore, an attention mechanism has also been applied for gaze tracking in CNN-based gaze tracking in [6]. All these works demonstrate the effectiveness of CNN for appearance-based gaze tracking.

### C. Gaze Tracking Data Sets

Several publicly available gaze data sets have been built for gaze estimation [5], [7], [10], [11], [45]–[47]. However, most of the earlier data sets [45], [46] only used the image captured by a single-view camera for gaze estimation. Thus, the depth between the eyes and the screen could not be well estimated, which reduced the accuracy of the gaze point estimation. Sugano *et al.* [10] synthesized images for arbitrary head poses, deployed a multiview cameras system to capture eye images, and built the UT Multiview data set. However, subjects needed to be fixed in a holder to watch the screen in their setting, which was in an unrealistic condition. Consequently, there were only 50 participants in this UT Multiview data set. Recently, Krafka *et al.* [5] utilized a crowdsourcing method to collect a very large gaze data set, including 1474 participants. In [5], it showed that more participants would boost the generalization of gaze tracking for the unseen participant. Because the number of participants in UT Multiview is too few, we propose to build a multiview gaze point estimation data set with more participants.

## III. SHANGHAITECHGAZE DATA SET

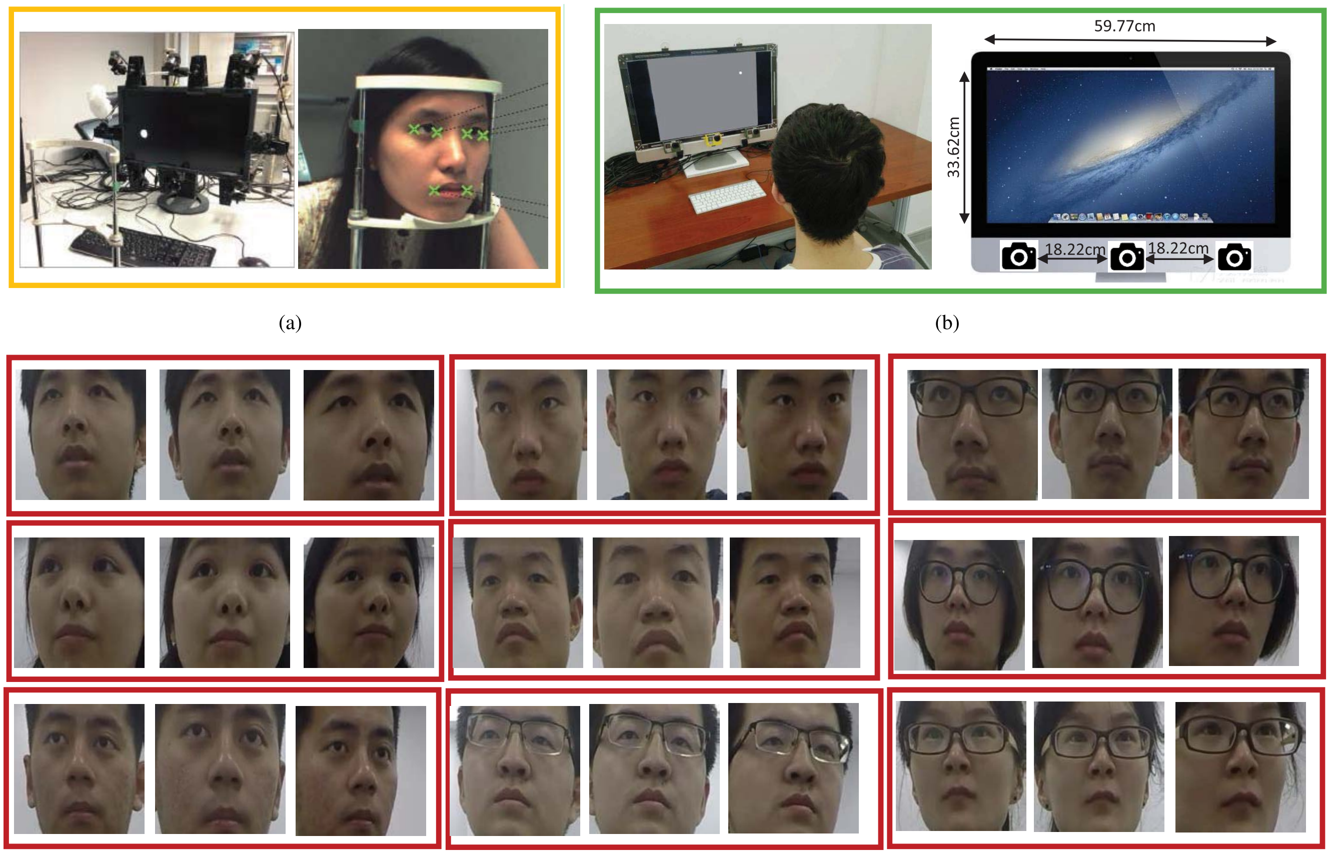
As aforementioned, the number of camera views [10] and the number of subjects [5] are the two key components for more accurate gaze point estimation. However, the number of subjects in the existing multiview gaze tracking data set (UT Multiview) [10] is too few (only 50 subjects). Even though the GazeCapture [5] data set is collected based on crowdsourcing and contains many subjects, it only utilizes one camera view. The single-view image lacks the important depth information, which limits the further improvement of performance. Therefore, we propose to build a multiview gazing tracking data set, which is named as the ShanghaiTechGaze data set. We list the statistics of the existing gaze tracking data sets and our data set in Table I. Compared with existing data sets, ShanghaiTechGaze has the following characteristics.

1) *It Is a Multiview Gaze Tracking Data Set:* Three cameras are deployed at the bottom of the screen to record the images of the eye and face in different views.

TABLE I

COMPARISON BETWEEN OUR DATA SET AND SOME PUBLICLY AVAILABLE GAZE TRACKING DATA SETS. WE USE THE FOLLOWING ABBREVIATIONS: CONT. FOR CONTINUOUS, ILLUM. FOR ILLUMINATION, AND COORD. FOR GAZE POINT COORDINATES

| Dataset | #Participants | #Poses | #Targets | #Views | Illum. | #Images | Angle/Coord. | Screen size (cm) |
|---|---|---|---|---|---|---|---|---|
| Eyediap [47] | 16 | cont. | cont. | 1 | 2 | videos | angle | 24” screen |
| OMEG [48] | 50 | 3 + cont. | 10 | 1 | 1 | 45,000 | angle | unknown |
| MPIIGaze [11] | 15 | cont. | cont. | 1 | cont. | 213,659 | angle | $28.65 \times 17.90$ (usually) |
| TabletGaze [7] | 51 | cont. | 35 | 1 | cont. | videos | coord. | $22.62 \times 14.14$ |
| GazeCapture [5] | 1474 | cont. | cont. | 1 | cont. | 2,445,504 | coord. | iphone or tablet |
| UT Multiview [10] | 50 | 8 + synth. | 160 | 8 | 1 | 64,000 | angle | $47.38 \times 29.61$ |
| Ours | 137 | cont. | cont. | 3 | cont. | 233,796 | coord. | $59.77 \times 33.62$ |




(a) (b)

(c)

Fig. 2. Images in the first row correspond to UT Multiview data acquisition system and ShanghaiTechGaze data acquisition system, respectively. Images in the second row correspond to faces captured by multiview cameras in our ShanghaiTechGaze data set. (a) Data acquisition system of UT Multiview. (b) Our data acquisition system. (c) Our multiview gaze tracking data set. Each group from left to right contains face images captured by left, middle, and right cameras.

2) *Images Are Captured Under an Uncontrolled Condition:* Different from [10], we do not place a chin holder during the experiment, and the participant can freely adjust the distance between his/her eyes and the screen, as well as the head pose.

## A. Our Gaze Tracking System

The gaze tracking system that we utilized for data acquisition is depicted in Fig. 2. Specifically, we use a 27-in Apple iMac machine as the display equipment. The width/height of the screen is 59.77 and 33.62 cm, respectively. Then, three GoPro Hero 4 cameras are deployed at the bottom of the screen to capture the image of a participant. The interval between two neighboring cameras is 18.22 cm. Then, the system is fixed on the desk in the room under a normal lighting condition. In order to avoid the disturbance caused by other moving objects/persons or noises, the room is kept quiet and empty except for a standby assistant. The first reason to use the large iMac is that we want to predict the gaze point with a larger range in both horizontal and vertical directions. In contrast, only points within the screen of the mobile phone or tablet are predicted for GazeCapture [5]. Thus, our data set is more challenging than GazeCapture. The second reason to use iMac is that its retina screen has a higher resolution to guarantee the pixelwise accuracy of the ground-truth; meanwhile, it reduces the eye fatigue during the data acquisition process.

### B. Data Acquisition Procedure

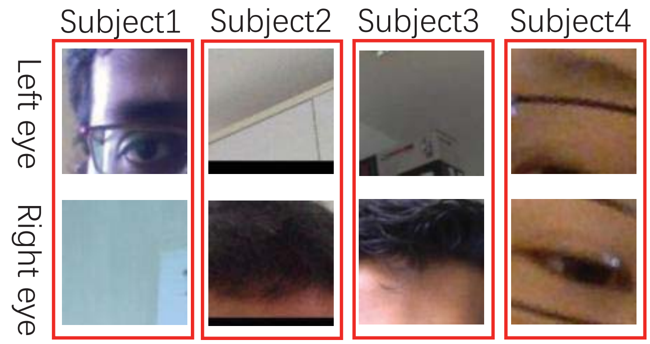
We ask a participant to sit in front of the screen freely. Then, we randomly display a white dot (as ground truth) whose radius is 8 pixels on the screen with the gray background and let the participant use a mouse to click the dot. Such a click action can help draw the participant's attention on this dot. Then, we record the coordinates of the cursor and time stamp of the action, after which a blue dot on the cursor position is displayed. Also, we calculate the distance between the white dot and the blue dot. If the distance exceeds a certain threshold (8 pixels in our setting), it is possible that the participant does not stare at the white dot, thus this data sample will be discarded. In our acquisition procedure, GoPro cameras are set to the video mode. Based on the time stamp of click action, we can extract the image frames of participants from the video.

For each participant, we first ask him/her to click nine dots to get familiar with the data acquisition system before recording data. Next, 50 dots will be displayed to the participant sequentially in each session for data acquisition. The screen flashes and the next dot will be displayed after the participant clicks the previous dot successfully. Each participant is asked to click dots for 12 sessions (click 600 dots in total). In between two sessions, we set a 1-min break to avoid the eye strain.

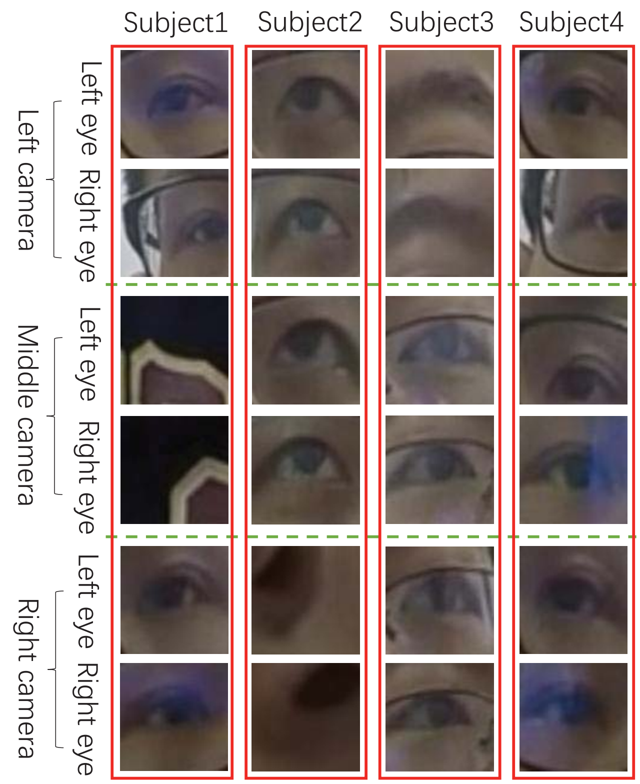
### C. Data Set Analysis

In total, we recruit 137 student participants for data collection (98 males and 39 females between 20 and 24 years old). All participants have normal or corrected-to-normal vision. After removing the data samples whose distance between the white dot and the blue dot is larger than the threshold, about 450–600 dots and their corresponding images of eyes and faces are kept for each participant. In the end, our ShanghaiTechGaze data set consists of 233 796 images. We further use the images corresponding to 100 participants as the training set and the images corresponding to the remaining 37 participants as the test set.

Fig. 3 shows that sometimes the eyes cannot be detected correctly in the MPIIGaze and our data set due to the limitation of the face detection algorithm, the complexity of the background, or the occlusion of the eyeglasses. The performance of gaze estimation extremely depends on the quality of the eye images. Although the detection algorithm cannot detect the eye images correctly in some view(s) or the detected eye images are low quality, in our data set, it is possible to detect (high quality) eye images in other views. Therefore, our solution is more robust for gaze tracking.




(a)




(b)

Fig. 3. Some eye images in MPIIGaze and our data set. It shows that multiview cameras would facilitate gaze estimation because it is more robust for data acquisition. (a) Some examples that both eyes cannot be detected correctly in MPIIGaze. (b) Even though eye images in some view(s) are not correctly detected or detected with low quality, eye images in another view(s) can be correctly detected or detected with high quality.

## IV. Method

### A. Relationship Between Gaze Direction Prediction and Gaze Point Prediction

Gaze direction prediction and gaze point prediction are two different but related tasks. Based on the gaze direction and eye positions in the 3-D coordinate system, we can infer the gaze point on the screen. However, even if there are two algorithms with the same gaze direction prediction errors, their gaze point prediction errors may be different. We illustrate this in Fig. 4. For conciseness, we only consider a single-eye case. In this figure, $O$ is the eye position, $g_d$ is the 3-D gaze direction, $\alpha$ is the gaze direction error, $P_0$ is the ground truth of the 2-D gaze point, and $P_1$ and $P_2$ are the two intersections between the gaze direction error sight and screen target. That is to say, when the gaze direction error is the same ($\alpha$), the gaze point error may be different. It is because every point of the intersection between the gaze direction error sight and screen target has a different distance to $P_0$ (in fact, the intersection is ellipse).

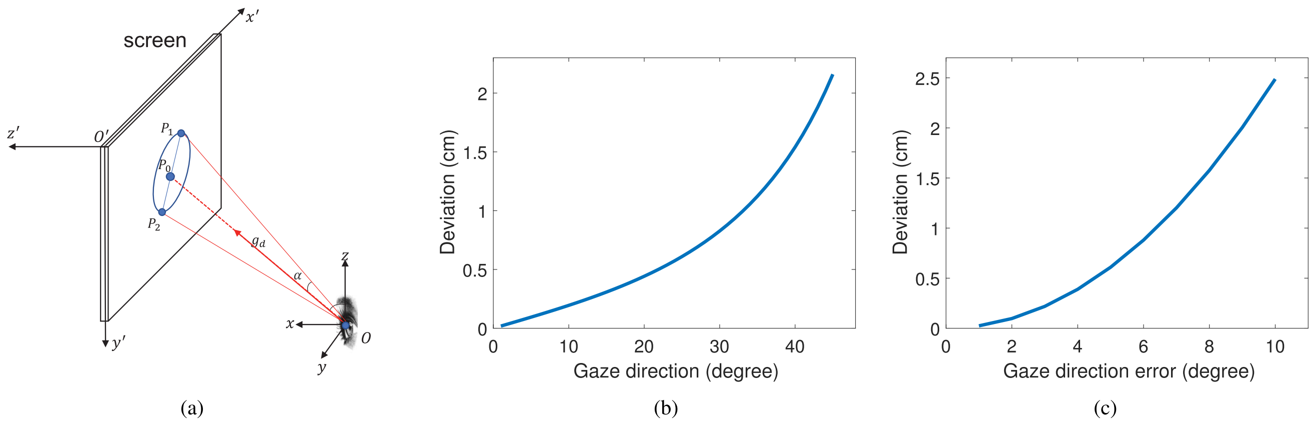



Fig. 4. Relationship between the gaze direction prediction and gaze point prediction. (a) Same gaze direction prediction errors may result in different gaze point errors. (b) When the vertical distance is 70 cm and the gaze direction error is 5°, the curve of gaze point prediction deviation with respect to the gaze direction. (c) When the vertical distance is 70 cm and the gaze direction is 25°, the curve of gaze point prediction deviation with respect to the gaze direction error.

For convenience, we assume that the largest deviation is $\|P_0P_1| - |P_0P_2\|$ (sometimes, the deviation is larger). This error cone is relevant to the vertical distance between the human eye and screen target, gaze direction, and gaze direction error. Assume that the vertical distance between the human eye and screen target is 70 cm, Fig. 4(b) shows the relationship between the deviation and gaze direction when the gaze direction error is 5°. Similarly, Fig. 4(c) shows the relationship between this deviation and gaze direction error when the gaze direction is 25°. It shows that even if the gaze direction errors of the two algorithms are the same, their gaze point errors may be very different.

Due to the geometry relationship between the gaze direction prediction and gaze point prediction, we propose to predict the gaze point based on the gaze direction and simultaneously minimize the prediction errors of both the gaze direction prediction task and gaze point prediction task within an MTL framework. We depict the network architecture used for our solution in Fig. 5.

### *B. Multitask Gaze Tracking for Single-View Image*

As aforementioned, the gaze direction estimation and gaze point estimation are closely related. Motivated by the success of MTL [13], [49], [50], it is desirable to extract features for both tasks and work on both tasks simultaneously. However, such gaze direction information is not available in our data set. Luckily, in MPIIGaze data set [11], such gaze direction information is provided. Thus, we propose to feed both the MPIIGaze data set and our ShanghaiTechGaze data set into a partially shared CNN, as shown in Fig. 5, and solve both problems within an MTL framework: 1) for images from MPIIGaze, their gaze direction vectors are predicted and 2) for images from ShanghaiTechGaze, their gaze point coordinates are estimated. Specifically, in Fig. 5, the whole network can be divided into two modules based on their functions: gaze direction and gaze point estimation modules. Next, we will detail these two modules separately.

*1) Module of Gaze Direction Estimation:* In the 3-D gaze direction estimation task, the ground truth is a 3-D unit vector whose elements are corresponding to the coordinates in a Cartesian coordinate system, and the initial points are the center of left eye and right eye, respectively. However, it is not proper to directly minimize the Euclidean distance between the ground truth and prediction because a small distance in Cartesian coordinate system does not guarantee the direction predicted close to the ground truth. Motivated by the work of Sugano *et al.* [10], we propose to change the Cartesian coordinate system to a spherical coordinate system and compress three variables to two variables. Mathematically, we denote $g^C(I_i) = (x, y, z)$ as a unit gaze direction vector in the Cartesian coordinate system corresponding to the gaze point in the $i$th image $I_i$, and denote $g^S(I_i) = (\theta, \phi, d)$ as its counterpart in the spherical coordinate system. The coordinate transformation equations are listed as follows:

$$\begin{cases} d = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arccos \dfrac{z}{r} \\ \phi = \arctan \dfrac{y}{x} \end{cases} \tag{1}$$

where $d$ is 1 because $g^C(I_i)$ is a unit vector. We denote the prediction of $g^S(I_i)$ as $\hat{g}^S(I_i)$, and the total number of gaze points with the gaze direction ground truth is $N$, then we arrive at the loss function corresponding to the gaze direction prediction $\ell_1$ (here, $I_i$ corresponds to the $i$th image from MPIIGaze)

$$\ell_1 = \frac{1}{N} \sum_{i=1}^{N} \left( \|g_l^S(I_i) - \hat{g}_l^S(I_i)\|_2^2 + \|g_r^S(I_i) - \hat{g}_r^S(I_i)\|_2^2 \right) \tag{2}$$

where $l$ and $r$ represent the gaze direction of the left and right eye, respectively.

*Coplanar Loss:* As shown in Fig. 1, $g_l^C$ and $g_r^C$ are the gaze direction of the left and right eye, respectively, and $v$ is the vector from the left eye to the right eye. Since both eyes are staring at the same gaze point $g_p$ on the screen, they satisfy a coplanar relation. Consequently, for the gaze directions of

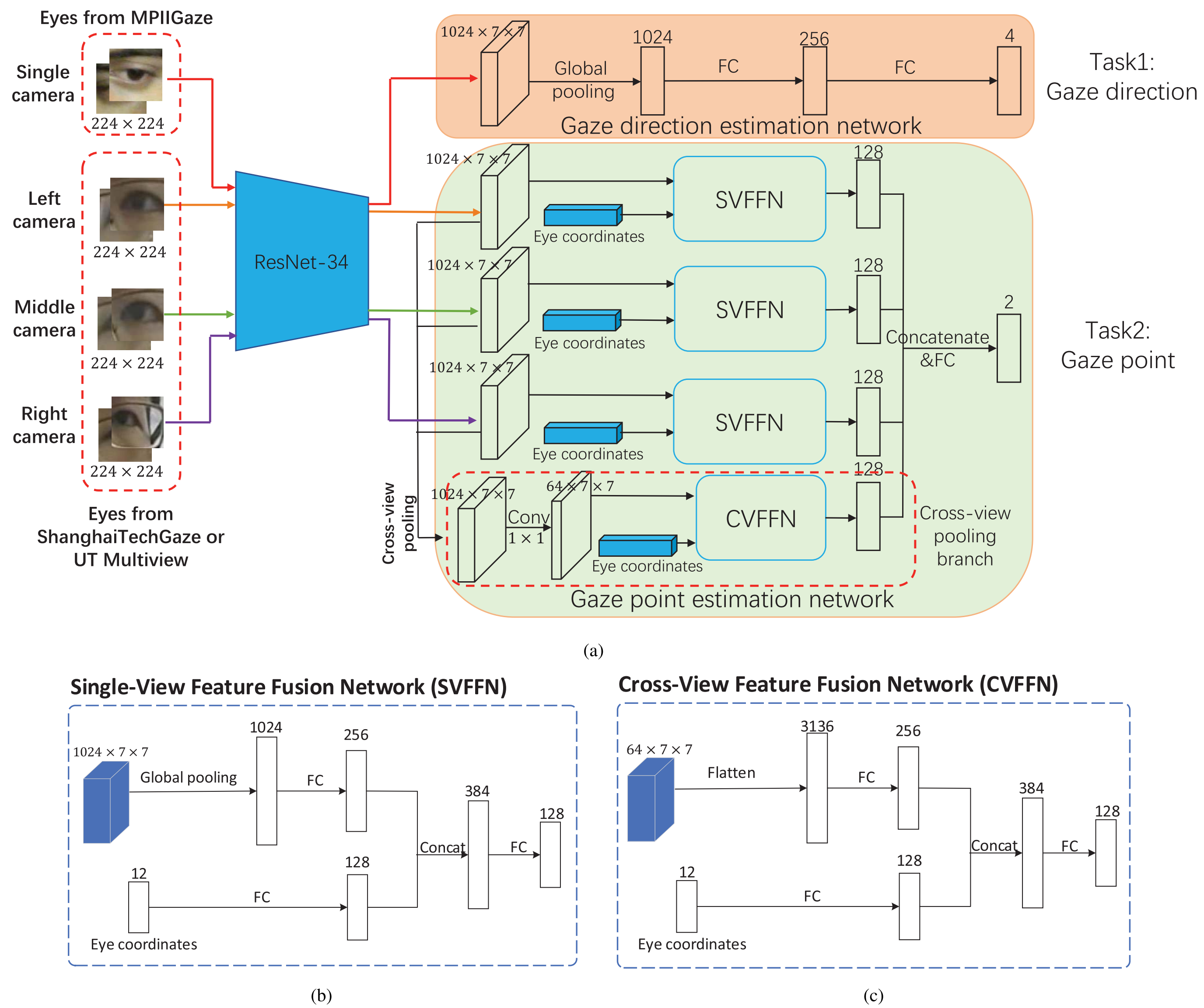



Fig. 5. Our multiview multitask network for the 3-D gaze direction and 2-D gaze point estimation. (a) Overall architecture of our network. (b) Specific details of the SVFFN module in (a). (c) Specific details of the CVFFN module in (a). As shown in (a), the two eyes images in MPIIGaze and six eyes images from three cameras in ShanghaiTech or UT Multiview are fed into an eye feature extractor (here, we use ResNet-34 [38]) for extracting features, whose outputs are eight feature maps with $512 \times 7 \times 7$. The feature maps for the same view are concatenated with size $1024 \times 7 \times 7$, then features from MPIIGaze are fed into the gaze direction estimation network and features from ShanghaiTechGaze and UT Multiview are fed into the gaze point estimation network.

both eyes in the $i$th image $I_i$, the following equation holds:

$$\ell_2 = \frac{1}{N}\sum_{i=1}^{N}\left\|g_l^C(I_i) \times g_r^C(I_i) \cdot v(I_i)\right\|^2 \tag{3}$$

where "$\times$" denotes the cross product of two vectors and "$\cdot$" denotes the inner product.

By combining the loss functions in (2) and (3), we arrive at the following loss function for the gaze direction estimation:

$$\begin{aligned}\ell_{\text{direction}} &= \ell_1 + \lambda_1 \ell_2 \\ &= \frac{1}{N}\sum_{i=1}^{N}\Big(\left\|g_l^S(I_i)-\hat{g}_l^S(I_i)\right\|_2^2+\left\|g_r^S(I_i)-\hat{g}_r^S(I_i)\right\|_2^2 \\ &\quad + \lambda_1\left\|g_l^C(I_i) \times g_r^C(I_i) \cdot v(I_i)\right\|^2\Big)\end{aligned} \tag{4}$$

where $\lambda_1$ is the weight of the coplanar loss.

*2) Module of Coordinate Estimation of Gaze Point:* Once we know the eye position in the given camera coordinate system, based on the gaze direction, we can predict the gaze point in the coordinate system of the screen. However, because the distance between the screen target and eyes (face of the participant) can vary arbitrarily, it is not easy to infer the eye position through the image captured by a single-view camera. Fortunately, we have three cameras deployed at the bottom of the screen. On the one hand, based on the coordinates of eyes in the image captured by these three cameras, we can infer the eye position in the world (camera) coordinate system. On the other hand, the coordinate transformation between the image coordinate system and the world (camera) coordinate system is fixed for all participants. Therefore, we can infer the eye position by using the coordinates of eyes in the images captured by all three cameras.

We denote the coordinates of the left and right eyes in the image captured by the $j$th ($j = 0, 1, 2$ stands for the left, middle, and right cameras, respectively) camera as $(x_l^j, y_l^j)$ and $(x_r^j, y_r^j)$, respectively. Then, we concatenate the coordinates of six eyes in these three cameras and get a 12-D coordinates vector $z = [x_l^0, y_l^0, \ldots, x_r^2, y_r^2]$. Such an eye coordinate vector encodes the information of the eye position. Next, we feed $z$ into our network, which is concatenated with the high-level features encoded by the gaze direction prediction information to predict the coordinates of the gaze point, as shown in Fig. 5.

We denote the gaze point in the $k$th image $I_k$ as $p(I_k)$, and denote its estimation as $\hat{p}(I_k)$. Suppose there are $M$ images with ground-truth gaze point coordinates. Then, we arrive at the following objective function (here, $I_k$ corresponds to the $k$th image from ShanghaiTechGaze or UT Multiview):

$$\ell_{\text{coordinate}} = \frac{1}{M}\sum_{k=1}^{M} \|p(I_k) - \hat{p}(I_k)\|_2^2. \tag{5}$$

*3) Loss Function in Multitask Gaze Tracking:* We combine the loss functions of the gaze direction estimation and gaze point estimation together and arrive at the following objective function:

$$\begin{aligned}\ell_{\text{single}} &= \ell_{\text{direction}} + \lambda_2 \ell_{\text{coordinate}} \\ &= \frac{1}{N}\sum_{i=1}^{N} \Big(\big\|g_l^S(I_i) - \hat{g}_l^S(I_i)\big\|_2^2 + \big\|g_r^S(I_i) - \hat{g}_r^S(I_i)\big\|_2^2 \\ &\qquad + \lambda_1 \big\|g_l^C(I_i) \times g_r^C(I_i) \cdot \upsilon(I_i)\big\|^2\Big) \\ &\quad + \lambda_2 \frac{1}{M}\sum_{k=1}^{M} \big\|p(I_k) - \hat{p}(I_k)\big\|_2^2 \end{aligned} \tag{6}$$

where $\lambda_2$ is the weight of $\ell_{\text{coordinate}}$.

## C. Multiview Multitask Gaze Tracking

We combine the images captured by cameras of different views and propose our multiview MTL framework. The whole network architecture is shown in Fig. 5. Specifically, we combine the features outputted by different views of subnetwork and fuse them with a fully connected layer for the gaze point estimation. Furthermore, we share the parameters of subnetworks from different views to reduce the number of parameters. Furthermore, motivated by the work of Su *et al.* [51], we also introduce a cross-view pooling layer to aggregate the features from different views, and such a cross-view pooling layer is useful if the detailed information is missing or some noises exist in some views, for example, the information loss caused by light reflection of eyeglasses. Then, we concatenate a high-level cross-view pooling feature with original single-view high-level features for the gaze point prediction. The overall objective function of our multiview multitask gaze tracking is shown as follows:

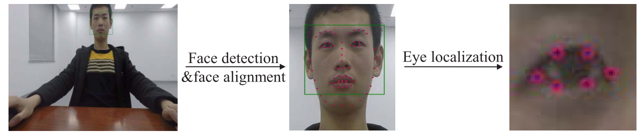



Fig. 6. Eye extraction and localization in the ShanghaiTechGaze data set.

$$\begin{aligned}\ell_{\text{multiple}} &= \frac{1}{N}\sum_{i=1}^{N} \Big(\big\|g_l^S(I_i) - \hat{g}_l^S(I_i)\big\|_2^2 + \big\|g_r^S(I_i) - \hat{g}_r^S(I_i)\big\|_2^2 \\ &\qquad + \lambda_1 \big\|g_l^C(I_i) \times g_r^C(I_i) \cdot \upsilon(I_i)\big\|^2\Big) \\ &\quad + \lambda_2 \frac{1}{M}\sum_{k=1}^{M} \big\|p\big(I_k^L, I_k^M, I_k^R\big) - \hat{p}\big(I_k^L, I_k^M, I_k^R\big)\big\|_2^2 \end{aligned} \tag{7}$$

where $I_k^L$, $I_k^M$, and $I_k^R$ corresponding to the images captured by left-, middle-, and right-view cameras for the $k$th gaze point in ShanghaiTechGaze or UT Multiview, and $I_i$ corresponds to images from MPIIGaze.

## D. Implementation Details

To maintain consistency with [11], we employ the same method as that in [11] to detect face and facial landmarks. The center of each eye is determined by averaging the six landmarks surrounding eyes, as shown in Fig. 6. Then, a square patch with a length of 1.5× the distance between eye corners is cropped. We then resize the patch to 224 × 224 pixels and feed it into a shared ResNet-34 [38] for feature extraction. The size of feature extracted by ResNet-34 is $512\times7\times7$, where the 512 is the number of channels in the feature maps, and $7\times7$ is the spatial resolution of the feature maps. Then, feature maps of both eyes in the same view are concatenated and followed by a global pooling. Then, we add two fully connected layers for the gaze direction prediction. The network architecture employed for gaze point prediction is similar to that in gaze direction prediction, as shown in Fig. 5. In addition to using features extracted from different views, we also conduct a cross-view pooling [51]. Eventually, we concatenate features corresponding to left, middle, and right views and cross-view pooling branch for 2-D gaze point prediction.

Motivated by the success of deeply supervised nets [52], we also add the middle layer supervision for an extracted feature from each view in the training of our multiview multitask network. For the MTL of two data sets, we set the ratio of the number of samples from ShanghaiTechGaze and that from MPIIGaze to 1: four for each minibatch. The parameters of the network are optimized based on the SGD optimization algorithm.

# V. Experiments

In this section, we empirically evaluate our method with our ShanghaiTechGaze multiview data set and the UT Multiview data set. The MPIIGaze data set is used for the gaze direction estimation in MTL. For UT Multiview, it provides images for eight views. To maintain consistency with the experimental setup in both data sets, only images corresponding to three cameras at the bottom of the screen are used.

### A. Experimental Setup

Our method is implemented based on the PyTorch framework. We train our network with the following hyperparameters: minibatch size (160 images of which 32 images are from ShanghaiTechGaze and 128 from MPIIGaze), learning rate ($10^{-5}$), momentum (0.9), weight decay (0.0005), and a number of epochs (80). In the training stage, we set the weight for coplanar loss term ($\lambda_1$) as $10^{-6}$ to make the value of $\ell_1$ term and $\ell_2$ term in (4) be at the similar magnitude level. Otherwise, the term with larger value would dominate the optimization process. We set the weight for the gaze point estimation term ($\lambda_2$) as $10^{-3}$.

*Evaluation Metrics:* Different evaluation metrics are adopted for different tasks. Specifically, we use the Euclidean distance between our predicted point and ground-truth point as the metric for the gaze point estimation evaluation, that is,

$$e_d = \frac{1}{M} \sum_{i=1}^{M} \| p_i - \hat{p}_i \|_2 \tag{8}$$

where $M$ is the number of total images, $p_i$ is the ground truth of the $i$th image, $\hat{p}_i$ is the predicted gaze point in the $i$th image, and $e_d$ is the averaged distance between the predicted gaze point and ground truth.

For the gaze direction estimation, the angle error between our estimated direction and ground-truth direction is used for evaluation, that is,

$$e_a = \frac{1}{N} \sum_{i=1}^{N} \arccos \frac{g_i \cdot \hat{g}_i}{|g_i \| \hat{g}_i|} \tag{9}$$

where $g_i$ is the ground truth of the $i$th image, $\hat{g}_i$ is the predicted gaze direction in the $i$th image, and $e_a$ is the averaged angle error between the predicted gaze direction and ground truth.[2]

In the following experiments, we use (8) for the 2-D gaze point estimation measurement (unit: centimeter) and (9) for the 3-D gaze direction estimation measurement (unit: degree).

### B. Performance Comparison

Since our method is based on deep learning, we compare our method with the following deep learning-based gaze tracking methods.

1) *Multimodal CNN [11]:* It feeds both eye images and head pose into a multimodal CNN to estimate the gaze point.
2) *Itracker [5]:* It feeds eye images, face, and face grid into different CNNs and combines all the high-level features with a fully connected layer for the gaze point estimation.
3) *Spatial Weights CNN [12]:* It only feeds the whole face image into the spatial weights CNN for the gaze point estimation.

All these baselines are designed for single-view gaze tracking. For fair comparison under multiview setting for the gaze point estimation, we concatenate features in the fully connected layers from all three views and use another fully connected layer to estimate the gaze point for each baseline method. It is worth noting that the network parameters are shared between the subnetworks corresponding to different views for all baselines, which is the same for our multiview model. Then, we term the adapted methods under multiview setting as multimodal CNN for multiview, iTracker for multiview, and spatial weights CNN for multiview, respectively. Besides, we replace the network in iTracker [5] with the ResNet-34 [38] architecture and term such baseline as iTracker for multiview (ResNet-34). We further extend iTracker and spatial weights CNN-based methods to multitask setting and denote these two baselines as iTracker for multitask (ResNet-34) and spatial weights CNN for multitask (ResNet-34).

In addition, we also design another two baselines by changing eye input or network architecture for performance comparison.

1) *Single-CNN for Multiview:* We stack six eyes from all three views together, which makes the input data to have 18 channels (each eye has three RGB color channels). Then, we feed the input into a ResNet-34 for feature extraction. After feature extraction, we combine the feature and eye coordinates with the same way as that in ours for the gaze point estimation. We term such a baseline as SingleCNN for multiview. It directly combines features from different views as the input of a ResNet-34 while our method combines features from different views rather than raw images.
2) *Multi-CNN for Multiview:* We stack the left and right eye images corresponding to the same view together, which makes the input data to have six channels for the left, middle, and right camera, respectively. These data and the eye coordinates are fed into three ResNet-34; here, the ResNet-34 for different views is the same. The features corresponding to different views are concatenated with the eye coordinates in the same way as that in ours for the gaze point estimation. We term such a baseline as multi-CNN for multiview. Compared with single-CNN for multiview, multi-CNN for multiview combines features from different views other than raw input images. Compared with our method, multi-CNN for multiview is still as a single-task CNN. Furthermore, since we introduce the cross-view pooling, we extract features for different eyes separately while multi-CNN for multiview combines both eyes in one view as the input of ResNet-34 for feature extraction.

For the MPIIGaze data set, we select data corresponding to 10 subjects as a training set and that for the remaining 5 subjects as a test set. For the UT Multiview data set, we select data corresponding to 36 subjects as a training set and use that corresponding to the remaining 14 subjects as a test set from 50 subjects in total. For multimodal CNN [11], we add

[2]We evaluate the performance for the left eye and right eye separately.

TABLE II
PERFORMANCE COMPARISON ON THE MPIIGAZE DATA SET (UNIT: DEGREE)

| Methods | left eye | right eye |
|---|---|---|
| Multi-modal CNN [11] | 6.62 | 6.57 |
| iTracker [5] | 6.02 | 6.04 |
| Spatial weights CNN [12] | 6.05 | 6.06 |
| iTracker (ResNet-34) | 5.29 | 5.31 |
| Multi-view Multi-task with UT Multiview | 4.73 | 4.72 |
| Multi-view Multi-task with ShanghaiTechGaze | 4.55 | 4.55 |

TABLE III
PERFORMANCE COMPARISON ON UT MULTIVIEW AND SHANGHAITECHGAZE (UNIT: CENTIMETERS)

| Methods | UT Multiview | ShanghaiTechGaze |
|---|---|---|
| Multi-modal CNN for Multi-view [11] | 5.54 | 6.33 |
| iTracker for Multi-view [5] | 5.01 | 5.80 |
| Spatial weights CNN for Multi-view [12] | 5.15 | 5.82 |
| iTracker for Multi-view (ResNet-34) | 4.39 | 5.08 |
| iTracker for Multi-task (ResNet-34) | 4.44 | 5.15 |
| Spatial weights CNN for Multi-task (ResNet-34) | 4.34 | 5.21 |
| SingleCNN for Multi-view (ResNet-34) | 4.71 | 5.32 |
| MultiCNN for Multi-view (ResNet-34) | 4.58 | 5.19 |
| Multi-view Multi-task with MPIIGaze | 3.89 | 4.61 |

coordinates of two eyes as an extra input for the gaze point estimation. In the UT Multiview data set, the whole face images are not provided. Therefore, we use the eye coordinates instead of face grid as input when evaluating iTracker [5], which is the same as that in our method. Again, we also replace the face with eye images when evaluating spatial weights CNN [12].

The experimental results of different methods are listed in Tables II and III. We can see that for both the gaze direction estimation on MPIIGaze in Table II and the gaze point estimation on ShanghaiTechGaze and UT Multiview in Table III, our method always outperforms all the baselines, which validates the effectiveness of our model for gaze tracking. Furthermore, we find that even with the same network architecture, our multiview multitask CNN obtains better performance than iTracker for multitask Resnet-34. There are two possible reasons. First, we employ the 12-D eye coordinates from all three views as input for gaze point prediction, which contain the 3-D spatial coordinate information of eyes, which helps gaze prediction. Second, the single-view camera often suffers from self-occlusion and noises caused by specularity and color distortion of eyeglasses. While multiview cameras have the wide field of view, cross-view pooling chooses a suitable feature from different views for the gaze point estimation, which partially reduces the effect of noises and occlusion, hence boosts the performance of the gaze point estimation.

### *C. Effect of Multitask Learning*

To quantitatively evaluate the effect of MTL, we first conduct the gaze point estimation task only by using our method for multiview images on both ShanghaiTechGaze and UT Multiview. The network architecture is the same as ours as shown in Fig. 5 except for removing the gaze direction estimation loss. Then, we conduct the gaze direction estimation task only in the MPIIGaze data set. Both two single-task networks are learned separately. Finally, we utilized MTL to regress the gaze direction and gaze point. The performance comparisons between the single task- and multiple tasks-based gaze tracking are listed in Table IV.

The results in Table IV show that MTL always outperforms single-task learning in both the gaze point estimation task and gaze direction estimation task. As we discussed previously, such two tasks are related to each other and can supervise the network to optimize parameters from two different domains. Improving the performance of the gaze point estimation task implicitly helps the parameter optimization in the gaze direction estimation task and so does it in turn. Even though we do not have data with both the gaze direction ground truth and gaze point ground truth, enforcing the data from different data sets share the same network, which helps the parameter optimization, as well as improves the performance of both tasks.

### *D. Importance of Multiview Cameras Based Gaze Tracking*

To validate the effectiveness of multiview data for gaze tracking, we also conduct experiments on single-view data and multiview data in our ShanghaiTechGaze data set, the UT Multiview data set, as well as the MPIIGaze data set. For all experiments, MTL is used. The performance comparisons between the multiview data-based gaze tracking and single-view data-based gaze tracking are listed in Table V. It shows that multiview eye images promote the performance of the gaze direction prediction marginally but contribute much to the improvement of gaze point prediction. The gaze point estimation depends on the gaze direction estimation and eye positions, which hints the effectiveness of multiview data for eye position's inference. We further show the performance of our method for images with eyeglasses and without eyeglasses, and we can see that gaze tracking for images without eyeglasses is easier because of the occlusion, specularity, and color distortion caused by eyeglasses. In the multiview setting, the improvement for eyes with glasses is higher than that for eyes without glasses, which validates the importance of multiview architecture for reducing the occlusion and noises of eye images.

### *E. Network Architecture Evaluation*

In order to obtain better network architecture, we implement the following experiments for the single-task gaze estimation on ShanghaiTechGaze and MPIIGaze.

*1) Face Grid Versus Eye Coordinates:* For the gaze point estimation, it is important to know the eye position. To this end, Krafka *et al.* [5] used a face grid to infer the eye position. Specifically, a face grid is expressed as a binary mask at the position of the face in the original image. Then, such a binary face grid is fed into a CNN as the eye positions. However, such an encoding method is complicated and not accurate enough. In this paper, we propose to encode the eye coordinates in three cameras as input and use a network with only one fully connected layer to estimate the eye position. Compared with the face grid-based method, our solution is more intuitive and

TABLE IV

MULTITASK VERSUS SINGLE TASK ON THE THREE DATA SETS. THE UNIT FOR MPIIGAZE IS DEGREE, AND THE UNIT FOR SHANGHAITECHGAZE AND UT MULTIVIEW IS CENTIMETER

| | MPIIGaze (+ UT Multiview) | | UT Multiview | MPIIGaze (+ ShanghaiTechGaze) | | ShanghaiTechGaze |
|---|---|---|---|---|---|---|
| | left eye | right eye | | left eye | right eye | |
| Single-task | 4.83 | 4.84 | 4.57 | 4.83 | 4.84 | 5.26 |
| Multi-task | 4.73 | 4.72 | 3.89 | 4.55 | 4.55 | 4.61 |

TABLE V

MULTIVIEW VERSUS SINGLE VIEW ON THE THREE DATA SETS. THE UNIT FOR MPIIGAZE IS DEGREE, AND THE UNIT FOR SHANGHAITECHGAZE AND UT MULTIVIEW IS CENTIMETER

| | left view | middle view | right view | multi-view |
|---|---|---|---|---|
| MPIIGaze (+ UT Multiview) left eye | 4.76 | 4.75 | 4.79 | 4.73 |
| MPIIGaze (+ UT Multiview) right eye | 4.77 | 4.76 | 4.77 | 4.72 |
| UT Multiview | 4.82 | 4.76 | 4.83 | 3.89 |
| MPIIGaze (+ ShanghaiTechGaze) left eye | 4.63 | 4.65 | 4.68 | 4.55 |
| MPIIGaze (+ ShanghaiTechGaze) right eye | 4.65 | 4.66 | 4.72 | 4.55 |
| ShanghaiTechGaze | 5.51 | 5.58 | 5.62 | 4.61 |
| ShanghaiTechGaze w/o. eyeglasses | 5.01 | 4.98 | 5.02 | 4.15 |
| ShanghaiTechGaze w. eyeglasses | 5.75 | 5.88 | 5.94 | 4.83 |

TABLE VI

EFFECT OF COPLANAR LOSS IN THE MPIIGAZE DATA SET WITH SINGLE TASK (UNIT: DEGREE)

| Methods | left eye | right eye |
|---|---|---|
| Without coplanar loss | 5.15 | 5.17 |
| With coplanar loss | 4.83 | 4.84 |

TABLE VII

PERFORMANCE COMPARISON WITH DIFFERENT FEATURE AGGREGATION STRATEGIES IN THE SHANGHAITECHGAZE DATA SET WITH SINGLE TASK (UNIT: CENTIMETERS)

| Pooling methods | All | w/o. eyeglasses | w. eyeglasses |
|---|---|---|---|
| Concatenation only | 5.52 | 4.94 | 5.79 |
| Cross-view pooling only | 5.43 | 4.88 | 5.70 |
| Cross-view pooling + Concatenation | 5.26 | 4.77 | 5.49 |

interpretable. We compare face grid-based gaze tracking and eye coordinates-based gaze tracking in the ShanghaiTechGaze data set. Our 12-D eye coordinates[3] (5.26 cm) is more effective than the face grid (5.61 cm). It shows that eye coordinates-based gaze tracking achieves better performance than face grid-based gaze tracking. Meanwhile, 12-D coordinates based on three cameras can capture the information of the eye position.

*2) Effect of Coplanar Loss:* Coplanar loss constrains that gaze directions of two eyes are on a plane. We evaluate the effectiveness of the coplanar loss in the MPIIGaze data set with our gaze direction estimation architecture. Table VI shows that the coplanar loss can effectively improve the gaze direction prediction.

*3) Cross-View Pooling Versus Naive Concatenation:* In Table VII, we list the performance of different feature aggregation strategies for features from different views. It shows that the cross-view pooling helps to enhance performance. Furthermore, we can see that cross-view pooling is more useful for subjects with eyeglasses because it may get rid of the reflection effect and color distortion effect caused by eyeglasses.

## *F. Effect of the Number of Participants*

One important reason motivating us to build a multiview gaze tracking data set is that the previous work [5] shows more participants in the single-view gaze point estimation help to

[3]12-D eye coordinates mean the center coordinates of six eyes in the images captured by multiview cameras.

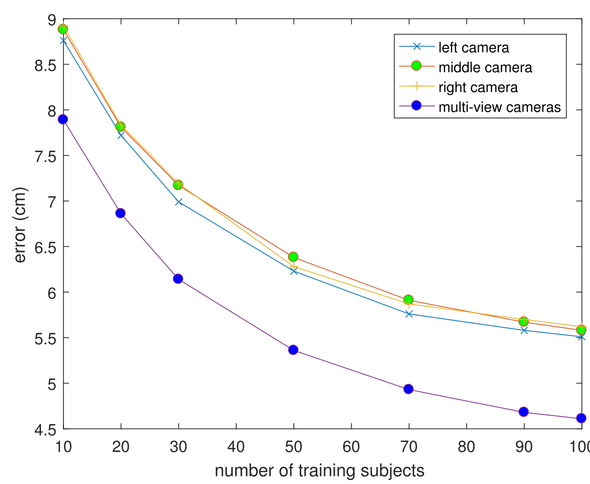



Fig. 7. Relationship between gaze point estimation error and the number of participants in the training set in our data set. The horizontal axis is the number of training subjects, and the vertical axis indicates the estimation error on the test set. It shows that more participants in the training set improve the generalization of gaze point estimation.

enhance the generalization ability of a deep learning model. However, the existing multiview gaze tracking data set, UT Multiview, only contains 50 participants. Our data set contains 137 participants, which is about three times larger than UT Multiview in terms of the number of participants. To validate the effect of the number of participants for multiview gaze

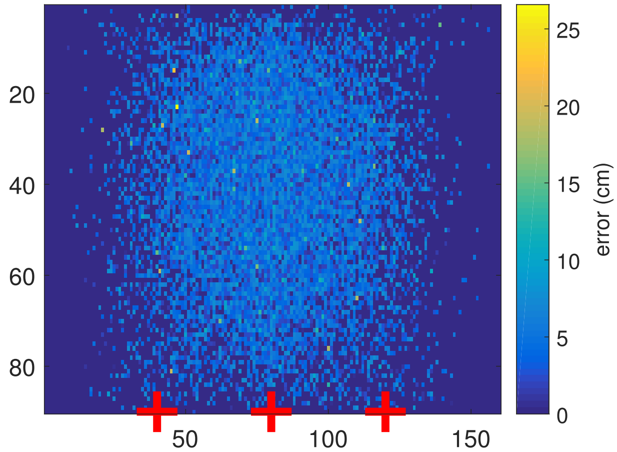



Fig. 8. Error distribution. Red crosses: cameras (best viewed in color).

TABLE VIII

PERFORMANCE AND RUNNING TIME FOR DIFFERENT BASELINE METHODS AND OUR METHOD WITH DIFFERENT NETWORK ARCHITECTURES

| Different methods and architectures | Performance (cm) | Running time (ms) |
|---|---|---|
| Multi-modal CNN for Multi-view [11] | 6.33 | 2.94 |
| iTracker for Multi-view [5] | 5.80 | 23.14 |
| Spatial weights CNN for Multi-view [12] | 5.82 | 39.22 |
| Multi-view Multi-task CNN with AlexNet | 4.99 | 15.79 |
| Multi-view Multi-task CNN with ResNet-34 | 4.61 | 21.34 |
| Multi-view Multi-task CNN with ResNet-50 | 4.44 | 36.25 |
| Multi-view Multi-task CNN with DenseNet-121 | 4.48 | 37.45 |

tracking setting, we fix the number of test data to 37 participants, which is the same as our previous experiments in our data set. Then, we use the data corresponding to a different number of participants to train the network. The performance of the network trained with a different number of participants is shown in Fig. 7, in which we can see that more participants gradually reduce the gaze point estimation error that validates the importance of more participants in the training set for multiview gaze tracking.

### G. Error Distribution

We also show the error distribution in our ShanghaiTechGaze data set. Following the work of Krafka *et al.* [5], we show the error in terms of the Euclidean distance at each ground-truth point. The results are shown in Fig. 8. It shows that points close to the center of the screen usually are with smaller errors.

### H. Time Cost

We compare the performance and time costs of both baseline methods and our method with different network architectures in the test phase and list the results in Table VIII. These results corresponding to networks are only for the multiview 2-D gaze estimation task in the ShanghaiTechGaze data set. All methods are tested on an NVIDIA K40 GPU. The performance represents the average distance error on the test data, and the running time represents the test time for one test sample. It is the average time executing 1000 iterations continuously. In terms of model selection, although the ResNet-50/DenseNet-121 [38], [53] leads to a higher gaze tracking accuracy, its running time is doubled. To balance the performance and running time, we choose ResNet-34 as our eye feature extractor. From Table VIII, we can see that multimodal CNN only contains two layers, therefore it is much faster. However, its performance is also worse than other baselines. Our method (ResNet-34) is faster than iTracker and spatial weights CNN baselines; meanwhile, our performance outperforms both, which validates the effectiveness of our method.

## VI. CONCLUSION

In this paper, we propose a multiview multitask framework to deal with both the gaze direction estimation task and gaze point estimation task simultaneously. To obtain more accurate eye positions and reduce the occlusion and noises in eye images, we utilize multiview cameras for the gaze point estimation. Since the only existing multiview gaze tracking data set, UT Multiview, is too small and the number of participants in that data set is too few, we build a new multiview gaze tracking data set, the ShanghaiTechGaze data set. Furthermore, motivated by the close relationship and difference between the gaze point estimation and gaze direction estimation, we propose a multitask solution to jointly solve these two tasks within a partially shared CNN. Extensive experiments on our ShanghaiTechGaze and the existing data sets validate the effectiveness of our multiview multitask gaze estimation framework.

## REFERENCES

[1] R. J. K. Jacob and K. S. Karn, "Eye tracking in human-computer interaction and usability research: Ready to deliver the promises," *Mind*, vol. 2, no. 3, p. 4, 2003.

[2] P. Majaranta and A. Bulling, "Eye tracking and eye-based human–computer interaction," in *Advances in Physiological Computing*. New York, NY, USA: Springer, 2014, pp. 39–65.

[3] C. H. Morimoto and M. R. M. Mimica, "Eye gaze tracking techniques for interactive applications," *Comput. Vis. Image Understand.*, vol. 98, no. 1, pp. 4–24, 2005.

[4] K. Rayner, "Eye movements in reading and information processing: 20 years of research," *Psychol. Bull.*, vol. 124, no. 3, p. 372, 1998.

[5] K. Krafka *et al.*, "Eye tracking for everyone," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 2176–2184.

[6] Y. Sugano, Y. Matsushita, and Y. Sato, "Appearance-based gaze estimation using visual saliency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 2, pp. 329–341, Feb. 2013.

[7] Q. Huang, A. Veeraraghavan, and A. Sabharwal. (Aug. 2015). "TabletGaze: Unconstrained appearance-based gaze estimation in mobile tablets." [Online]. Available: https://arxiv.org/abs/1508.01244

[8] C. Xu, D. Tao, and C. Xu. (2013). "A survey on multi-view learning." [Online]. Available: https://arxiv.org/abs/1304.5634

[9] J. Zhao, X. Xie, X. Xu, and S. Sun, "Multi-view learning overview: Recent progress and new challenges," *Inf. Fusion*, vol. 38, pp. 43–54, Nov. 2017.

[10] Y. Sugano, Y. Matsushita, and Y. Sato, "Learning-by-synthesis for appearance-based 3d gaze estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1821–1828.

[11] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling, "Appearance-based gaze estimation in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 4511–4520.

[12] X. Zhang, Y. Sugano, M. Fritz, and A. Bulling, "It's written all over your face: Full-face appearance-based gaze estimation," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jul. 2017, pp. 2299–2308.

[13] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 1440–1448.

[14] C. R. Qi, H. Su, M. Niessner, A. Dai, M. Yan, and L. J. Guibas, "Volumetric and multi-view cnns for object classification on 3D data," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 5648–5656.

[15] L. Ge, H. Liang, J. Yuan, and D. Thalmann, "Robust 3D hand pose estimation in single depth images: From single-view CNN to multi-view CNNs," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3593–3601.

[16] S. Ruder. (2017). "An overview of multi-task learning in deep neural networks." [Online]. Available: https://arxiv.org/abs/1706.05098

[17] Y. Zhang and Q. Yang. (2017). "A survey on multi-task learning." [Online]. Available: https://arxiv.org/abs/1707.08114

[18] S. Baluja and D. Pomerleau, "Non-intrusive gaze tracking using artificial neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 1994, pp. 753–760.

[19] K.-H. Tan, D. J. Kriegman, and N. Ahuja, "Appearance-based eye gaze estimation," in *Proc. 6th IEEE Workshop Appl. Comput. Vis. (WACV)*, Dec. 2002, pp. 191–195.

[20] R. Valenti, N. Sebe, and T. Gevers, "Combining head pose and eye location information for gaze estimation," *IEEE Trans. Image Process.*, vol. 21, no. 2, pp. 802–815, Feb. 2012.

[21] O. Williams, A. Blake, and R. Cipolla, "Sparse and semi-supervised visual mapping with the $S^3$GP," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 1, Jun. 2006, pp. 230–237.

[22] F. Lu, T. Okabe, Y. Sugano, and Y. Sato, "Learning gaze biases with head motion for head pose-free gaze estimation," *Image Vis. Comput.*, vol. 32, no. 3, pp. 169–179, 2014.

[23] K. A. F. Mora and J.-M. Odobez, "Person independent 3D gaze estimation from remote RGB-D cameras," in *Proc. 20th IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2013, pp. 2787–2791.

[24] E. Wood, T. Baltrusaitis, X. Zhang, Y. Sugano, P. Robinson, and A. Bulling, "Rendering of eyes for eye-shape registration and gaze estimation," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2015, pp. 3756–3764.

[25] D. W. Hansen and Q. Ji, "In the eye of the beholder: A survey of models for eyes and gaze," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 3, pp. 478–500, Mar. 2010.

[26] T. Ohno, N. Mukawa, and A. Yoshikawa, "FreeGaze: A gaze tracking system for everyday gaze interaction," in *Proc. Symp. Eye Tracking Res. Appl.*, 2002, pp. 125–132.

[27] D. H. Yoo and M. J. Chung, "A novel non-intrusive eye gaze estimation using cross-ratio under large head motion," *Comput. Vis. Image Understand.*, vol. 98, no. 1, pp. 25–51, 2005.

[28] Z. Zhu and Q. Ji, "Eye gaze tracking under natural head movements," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 918–923.

[29] Z. Zhu, Q. Ji, and K. P. Bennett, "Nonlinear eye gaze mapping function estimation via support vector regression," in *Proc. 18th Int. Conf. Pattern Recognit. (ICPR)*, vol. 1, Aug. 2006, pp. 1132–1135.

[30] C. Hennessey, B. Noureddin, and P. Lawrence, "A single camera eye-gaze tracking system with free head motion," in *Proc. Symp. Eye Tracking Res. Appl.*, 2006, pp. 87–94.

[31] T. Ishikawa, "Passive driver gaze tracking with active appearance models," in *Proc. 11th World Congr. Intell. Transp. Syst.*, 2004, pp. 41–43.

[32] J. Chen and Q. Ji, "3D gaze estimation with a single camera without ir illumination," in *Proc. 19th Int. Conf. Pattern Recognit. (ICPR)*, 2008, pp. 1–4.

[33] D. W. Hansen and A. E. Pece, "Eye tracking in the wild," *Comput. Vis. Image Understand.*, vol. 98, no. 1, pp. 155–181, 2005.

[34] W. Sewell and O. Komogortsev, "Real-time eye gaze tracking with an unmodified commodity Webcam employing a neural network," in *Proc. Extended Abstr. Hum. Factors Comput. Syst. (CHI)*, 2010, pp. 3739–3744.

[35] F. Lu, Y. Sugano, T. Okabe, and Y. Sato, "Adaptive linear regression for appearance-based gaze estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 10, pp. 2033–2046, Oct. 2014.

[36] F. Lu, Y. Sugano, T. Okabe, and Y. Sato, "Inferring human gaze from appearance via adaptive linear regression," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Nov. 2011, pp. 153–160.

[37] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.

[38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[39] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 3431–3440.

[40] H. Zhu, F. Meng, J. Cai, and S. Lu, "Beyond pixels: A comprehensive survey from bottom-up to semantic image segmentation and cosegmentation," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 12–27, Jan. 2016.

[41] L. Wang *et al.*, "Temporal segment networks: Towards good practices for deep action recognition," in *Proc. Eur. Conf. Comput. Vis.* New York, NY, USA: Springer, 2016, pp. 20–36.

[42] H. Zhu *et al.*, "YoTube: Searching action proposal via recurrent and static regression networks," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2609–2622, Jun. 2018.

[43] J. Yang, K. Yu, Y. Gong, and T. Huang, "Linear spatial pyramid matching using sparse coding for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2009, pp. 1794–1801.

[44] X. Peng, H. Tang, L. Zhang, Z. Yi, and S. Xiao, "A unified framework for representation-based subspace clustering of out-of-sample and large-scale data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 12, pp. 2499–2512, Dec. 2016.

[45] U. Weidenbacher, G. Layher, P.-M. Strauss, and H. Neumann, "A comprehensive head pose and gaze database," in *Proc. 3rd IET Int. Conf. Intell. Environ.*, Sep. 2007, pp. 455–458.

[46] B. A. Smith, Q. Yin, S. K. Feiner, and S. K. Nayar, "Gaze locking: Passive eye contact detection for human-object interaction," in *Proc. 26th Annu. ACM Symp. User Interface Softw. Technol.*, 2013, pp. 271–280.

[47] K. A. F. Mora, F. Monay, and J.-M. Odobez, "EYEDIAP: A database for the development and evaluation of gaze estimation algorithms from RGB and RGB-D cameras," in *Proc. Symp. Eye Tracking Res. Appl.*, 2014, pp. 255–258.

[48] Q. He *et al.*, "OMEG: Oulu multi-pose eye gaze dataset," in *Proc. Scand. Conf. Image Anal.* New York, NY, USA: Springer, 2015, pp. 418–427.

[49] R. Collobert and J. Weston, "A unified architecture for natural language processing: Deep neural networks with multitask learning," in *Proc. 25th Int. Conf. Mach. Learn.*, 2008, pp. 160–167.

[50] L. Deng, G. Hinton, and B. Kingsbury, "New types of deep neural network learning for speech recognition and related applications: An overview," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2013, pp. 8599–8603.

[51] H. Su, S. Maji, E. Kalogerakis, and E. Learned-Miller, "Multi-view convolutional neural networks for 3D shape recognition," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 945–953.

[52] C.-Y. Lee, S. Xie, P. Gallagher, Z. Zhang, and Z. Tu, "Deeply-supervised nets," in *Proc. Artif. Intell. Statist.*, 2015, pp. 562–570.

[53] G. Huang, Z. Liu, K. Q. Weinberger, and L. van der Maaten. (2016). "Densely connected convolutional networks." [Online]. Available: https://arxiv.org/abs/1608.06993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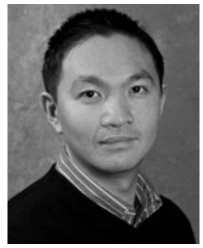

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